THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 13 & 14: **Statistics, Probability** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 6	$\frac{30}{5}$
2	**(a)** 35	Mid-value
3	**(a)** Mode = 3 Median − 2 Mean	Standard relation
4	**(a)** Incorrect — it is the class with the highest frequency	Definition
5	**(b)** Student A is correct	$\bar{x} + 5$
6	**(a)** $\frac{1}{2}$	1 of 2 outcomes
7	**(a)** $\frac{1}{2}$	$\frac{26}{52}$
8	**(d)** $\frac{1}{3}$	$\{3, 6\}$
9	**(a)** Incorrect — the sum is 1	Complementary events
10	**(b)** Student A is correct	Total 10

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{20+90+250}{20} = 18$	1+1
12	$20 + \frac{8-3}{16-3-5} \times 10 = 26.25$	1+1
13	$\frac{12}{20} = \frac{3}{5}$; $\frac{2}{5}$	1+1
14	HT, TH, TT ⇒ $\frac{3}{4}$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$n = 50$; cf 6, 16, 31 ⇒ $40 + \frac{25-16}{15} \times 20 = 52$	1+1+1
16	(i) $\frac{6}{36} = \frac{1}{6}$ (ii) $\frac{3}{36} = \frac{1}{12}$ (iii) $\frac{1}{6}$	1+1+1
17	$a = 60, h = 20$; $\Sigma fu = -16 - 12 + 0 + 6 + 8 = -14$ ⇒ $60 - \frac{280}{50} = 54.4$	1+1+1
	OR: $20 + \frac{9-5}{18-5-7} \times 10 = 26.67$	1+1+1
18	$\frac{13}{25}$, $\frac{8}{25}$, $\frac{9}{25}$	1+1+1
	OR: $\frac{x}{10+x} = \frac{1}{3} \Rightarrow x = 5$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1920}{60} = 32$ (ii) cf 4, 12, 24 ⇒ $30 + \frac{30-24}{20} \times 10 = 33$ (iii) $3(33) - 2(32) = 35$	2+2+1
	OR: $\frac{940+35p}{44+p} = 25 \Rightarrow p = 16$ (3). $n = 60$; cf 5, 23 ⇒ $20 + \frac{30-23}{15} \times 10 = 24.67$ (2)	3+2
20	(i) $\frac{12}{500} = \frac{3}{125}$ (ii) $\frac{1}{250}$ (iii) $\frac{122}{125}$ (iv) Wrong — the outcomes are not equally likely	1+1+2+1
	OR: (i) $\frac{1}{2}$ (ii) $\frac{27}{36} = \frac{3}{4}$ (iii) sums 4, 9: $\frac{7}{36}$ (iv) $\frac{3}{36} = \frac{1}{12}$	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 13 & 14: **Statistics, Probability** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 6	$\frac{2+4+6+8+10}{5}$
2	**(d)** 25	Larger end
3	**(a)** 13	$45 - 32$
4	**(b)** Correct	All frequencies included
5	**(b)** Student A is correct	$\bar{x} = a + 0$
6	**(d)** $\frac{1}{6}$	1 of 6
7	**(d)** $\frac{2}{11}$	2 B's in 11 letters
8	**(a)** $\frac{3}{13}$	$\frac{12}{52}$
9	**(a)** Incorrect — only $P(E) = 1$ means certain	0.6 < 1
10	**(a)** Student A is correct	Maximum sum is 12

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{6+8+18+16}{10} = 4.8$	1+1
12	$n = 40$; cf 4, 11, 26 ⇒ median class 10–15; modal class 10–15	1+1
13	$\frac{5}{6}$; $\frac{1}{3}$	1+1
14	$\frac{9}{15} = \frac{3}{5}$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a = 25$; $\Sigma fd = -120 - 80 + 0 + 90 + 140 = 30$ ⇒ $25 + \frac{30}{40} = 25.75$	1+1+1
16	(i) $\frac{45}{50} = \frac{9}{10}$ (ii) $\frac{5}{47}$	2+1
17	cf 5, 14, 28 ⇒ $30 + \frac{20-14}{14} \times 10 = 34.29$	1+1+1
	OR: $40 + \frac{13-8}{26-8-7} \times 20 = 49.09$	1+1+1
18	48 cards: $\frac{4}{48} = \frac{1}{12}$; $\frac{24}{48} = \frac{1}{2}$; 0	1+1+1
	OR: Multiples of 4: 7, of 6: 5, of 12: 2 ⇒ $\frac{10}{30} = \frac{1}{3}$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Median: cf 6, 16 ⇒ $200 + \frac{25-16}{15} \times 50 = 230$. Mode: $200 + \frac{15-10}{30-10-12} \times 50 = 231.25$	3+2
	OR: $\Sigma f = 44 + f$, $\Sigma fx = 752 + 20f$ ⇒ $\frac{752+20f}{44+f} = 18 \Rightarrow f = 20$ (4); modal class 19–21 (1)	4+1
20	(i) $\frac{3}{8}$ (ii) $\frac{15}{40} = \frac{3}{8}$ (iii) $\frac{5}{8}$ (iv) Wrong — sum is $\frac{5}{8}$; 'neither' also has probability	1+1+2+1
	OR: (i) $\frac{15}{36} = \frac{5}{12}$ (ii) $\frac{1}{6}$ (iii) $\frac{10}{36} = \frac{5}{18}$	2+1+2

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ANSWER KEY & MARKING SCHEME — SET C

Chapters 13 & 14: **Statistics, Probability** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 3	Most frequent
2	**(b)** class width	Step size
3	**(a)** 28	$\frac{24+60}{3}$
4	**(a)** Incorrect — it is approximate (frequencies assumed at class marks)	Assumption
5	**(b)** Student A is correct	$\frac{n}{2} = 25$ first reached at 37
6	**(d)** $\frac{1}{3}$	$\{3, 5\}$
7	**(c)** $\frac{1}{8}$	1 of 8
8	**(a)** $\frac{5}{9}$	E, U, A, I, O
9	**(c)** Incorrect — they come closer as trials increase	Concept
10	**(a)** Student A is correct	$\frac{2}{52}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{60+80+300}{20} = 22$	1+1
12	$30 + \frac{15}{60-15-18} \times 10 = 35.56$	1+1
13	$\{2, 3, 4, 5\} \Rightarrow \frac{2}{3}$	2
14	$\frac{32}{40} = \frac{4}{5}$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$n = 40$; cf 8, 20 ⇒ class 10–20: $10 + \frac{20-8}{12} \times 10 = 20$	1+1+1
16	$\frac{3}{26}$; $\frac{16}{52} = \frac{4}{13}$; $\frac{3}{4}$	1+1+1
17	$a = 250, h = 100$; $\Sigma fu = 6 \Rightarrow 250 + \frac{600}{40} = 265$	1+1+1
	OR: $5 + \frac{15-9}{30-9-9} \times 2 = 6$ — the student is correct	1+1+1
18	$\frac{1}{6}$; $\frac{5}{36}$; $\frac{11}{36}$	1+1+1
	OR: $\frac{7}{12}$; $\frac{6}{11}$	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $a = 75$: $\Sigma fu = 14 \Rightarrow 77.33$ h (ii) cf 6, 16, 34 ⇒ $70 + \frac{14}{18} \times 10 = 77.78$ h (iii) $70 + \frac{8}{36-26} \times 10 = 78$ h	2+2+1
	OR: $f_1 + f_2 = 9$; $32.5 = 30 + \frac{20-(f_1+14)}{12} \times 10 \Rightarrow f_1 = 3,\ f_2 = 6$	3+2
20	(i) $\frac{8}{20} = \frac{2}{5}$ (ii) $\frac{3}{5}$ (iii) adds only 23: $\frac{9}{25}$ (iv) No — P(win) < $\frac{1}{2}$	1+1+2+1
	OR: (i) 8 outcomes (ii) $\frac{3}{8}$ (iii) $\frac{4}{8} = \frac{1}{2}$ (iv) $\frac{1}{8}$	2+1+1+1

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ANSWER KEY & MARKING SCHEME SET D

Chapters 13 & 14: **Statistics, Probability** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 7	Middle value
2	**(c)** 10	$20 - 10$
3	**(c)** 24	$\frac{3(26)-30}{2}$
4	**(b)** Incorrect — both give the same mean	Only arithmetic is simplified
5	**(a)** Student A is correct	$\frac{0+20}{2}$
6	**(d)** 0	Impossible
7	**(a)** $\frac{10}{13}$	$\frac{40}{52}$
8	**(d)** $\frac{1}{4}$	Only TT
9	**(c)** Incorrect — it is $\frac{3}{4}$	HH, HT, TH
10	**(b)** Student A is correct	$\frac{5}{10}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{9+45+30}{10} = 8.4$	1+1
12	$30 + \frac{10-4}{20-4-6} \times 10 = 36$	1+1
13	2, 3, 5, 7 ⇒ $\frac{2}{5}$	2
14	$\frac{4}{11}$ each	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$n = 30$; cf 3, 10 ⇒ $150 + \frac{15-10}{10} \times 5 = 152.5$ cm	1+1+1
16	$\frac{1}{36}$; $\frac{1}{2}$; $\frac{4}{36} = \frac{1}{9}$	1+1+1
17	$a = 50, h = 10$; $\Sigma fu = -15$ ⇒ $50 - 5 = 45$ years	1+1+1
	OR: $7 + \frac{40-30}{80-30-16} \times 3 = 7.88$	1+1+1
18	Blue 18, green 24 ⇒ white 12	1+1+1
	OR: 16, 25, 36, 49 ⇒ $\frac{4}{50} = \frac{2}{25}$; 10 numbers ⇒ $\frac{1}{5}$	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mean $= \frac{2500}{50} = 50$ (2); median: cf 7, 17 ⇒ $40 + \frac{25-17}{15} \times 20 = 50.67$ (3)	2+3
	OR: (i) $\frac{1940}{50} = 38.8$ (ii) Wrong — frequencies ignored (iii) cf 12, 30 ⇒ $20 + \frac{13}{18} \times 20 = 34.44$	2+1+2
20	(i) $\frac{2}{5}$ (ii) $\frac{1}{10}$ (iii) $\frac{7}{10}$ (iv) $\frac{15}{30} = \frac{1}{2}$	1+1+1+2
	OR: (i) $\frac{1}{4}$ (ii) $\frac{1}{2}$ (iii) $\frac{1}{4}$ (iv) sum $= 1$ ✓ (v) Wrong — one head is twice as likely	1+1+1+1+1

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